Usability Evaluation of the XJTLU Museum AI System

Task success, usability, and visitor barriers in a short museum session

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01 PROBLEM & TARGET USERS

A public-facing Museum AI guide must work without a manual. Visitors may not know how to start, what to ask, where to go next, or how to recover from an unclear answer.

TARGET USERS

XJTLU students and staff, 18+, with mixed digital confidence and museum experience.

02 RESEARCH QUESTIONS

RQ1 Can visitors complete the four core Museum AI tasks?

RQ2 How usable and efficient is the system according to task performance and SUS?

RQ3 Which onboarding, navigation, answer, or recovery barriers recur?

H1: At least 80% independent task success.
H2: SUS reaches the benchmark of 68.
H3: Ease and confidence relate positively to SUS.
H4: Onboarding, navigation, clarity, or recovery barriers recur.

03 STUDY DESIGN

- Moderated task-based usability test
- 12-15 XJTLU students or staff
- One 15-minute session per participant
- Think-aloud comments and short debrief

04 FOUR VISITOR TASKS

TASK 1

Start

Open the guide and select an exhibit.

TASK 2

Ask

Ask one exhibit question and identify the answer.

TASK 3

Navigate

Move to the next recommended stop.

TASK 4

Recover

Resolve an unclear or unsuitable answer.

05 PROCEDURE & QUESTIONNAIRE

- Consent
- Warm-up
- Four tasks
- SUS
- Debrief

Questionnaire flow

Background · task completion and barriers · task ease/confidence/content usefulness · SUS · system-level clarity, navigation, recovery, trust and cognitive load · open comments.

Live bilingual survey: v.wjx.cn/vm/QZpCC3h.aspx

06 MEASURES

Observed

Task success, time on task, errors, moderator help

Self-report

SUS, ease, confidence, content usefulness, trust

Qualitative

Think-aloud hesitation, workarounds, open comments

07 PLANNED EVIDENCE

PENDING PARTICIPANT DATA

Insert the actual task-success profile, time and error counts, SUS score, system ratings, and anonymised comments after the pilot and main sessions.

08 INTERPRETATION & LIMITS

Connect each result to onboarding, exhibit content, navigation, recovery, or trust. The sample is small and the session is controlled. The study evaluates one system version and selected tasks.

09 RECOMMENDATIONS

- Clarify how to start the guide.
- Improve route feedback between exhibits.
- Add retry and correction paths for unclear answers.
- Keep the original label available and disclose AI assistance.

10 REFERENCES

XJTLU Museum, Technology and AI systems, 2026.
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J. Brooke, SUS: A quick and dirty usability scale, 1996.
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